Three Project Ideas

TECHIN 515

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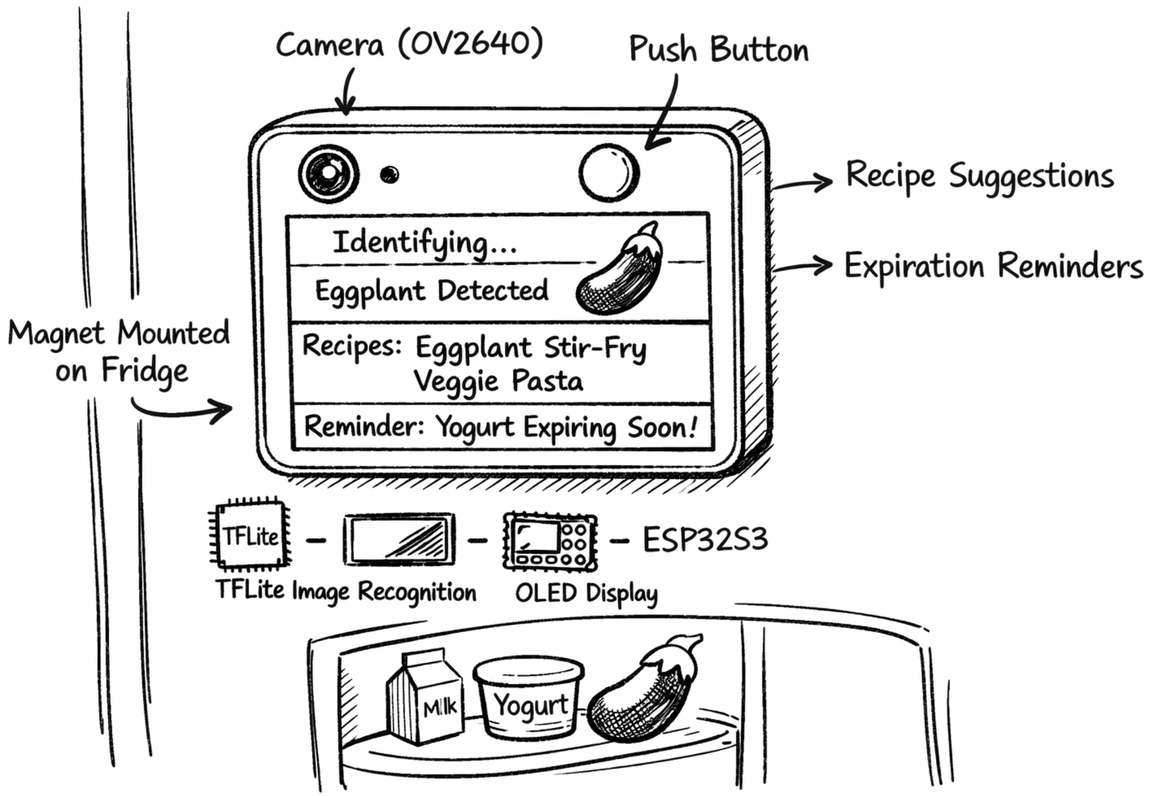
Fridge Ingredient Recognition Device

- A magnet-mounted camera device that attaches to the fridge door. When the user presses a button, the device captures and identifies the ingredient and suggests recipes based on what is available in the fridge, and reminds items that may be expiring soon.

- Intended user: People who cook at home regularly and want help deciding what to make without wasting food.

- Technical elements: A TFLite image classification model runs on the XIAO ESP32S3 for on-device ingredient recognition. Sensors include an OV2640 camera module and a pushbutton, with results displayed on an OLED screen.

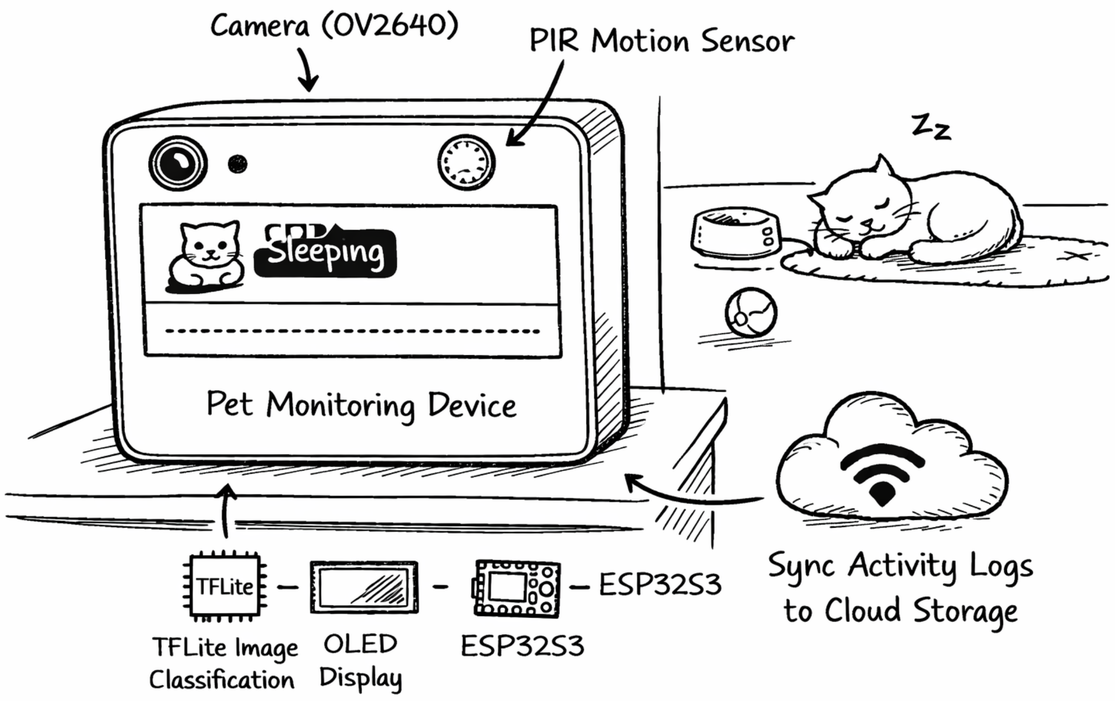


Pet Activity Monitor

- A compact camera device placed at home that continuously monitors your pet and classifies their current activity, such as sleeping, playing, or eating. The device processes everything on-board and shows the current activity state on a small display, with activity logs synced to cloud storage for owners to review later.

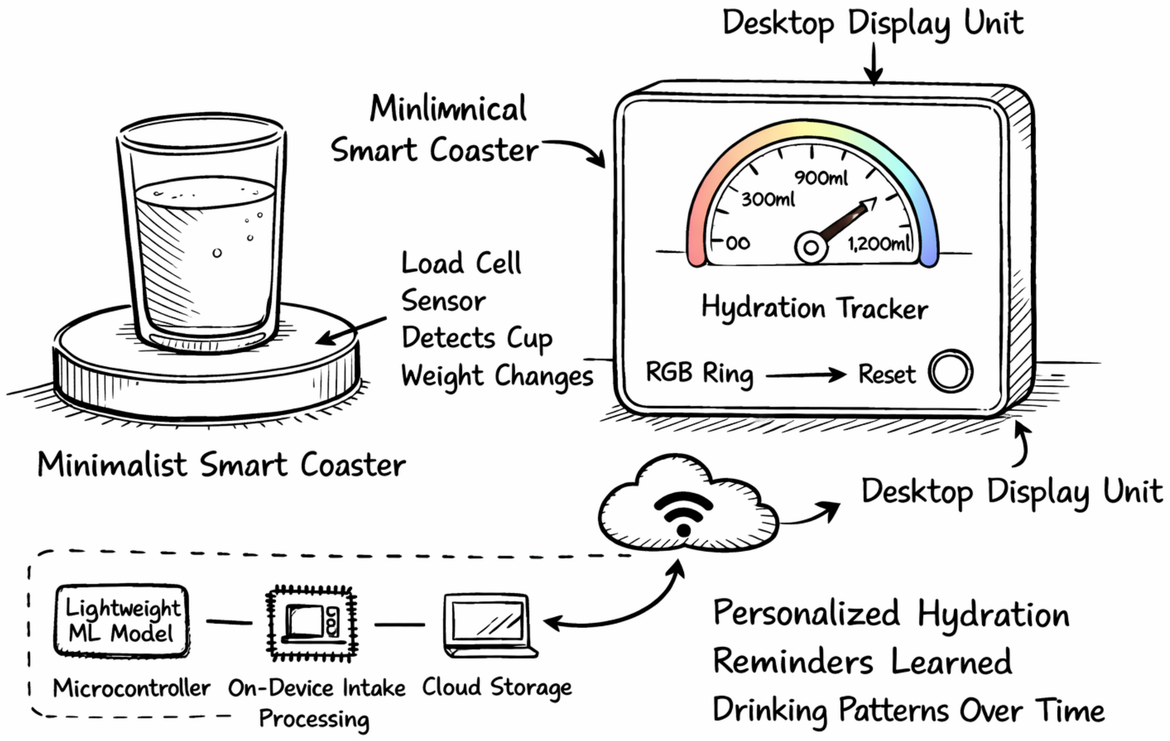
- Intended user: Pet owners who want to keep an eye on their cat or dog while away from home, or simply understand their pet's daily routine better.

- Technical elements: A TFLite image classification model runs on the XIAO ESP32S3 to recognize pet activity states from the OV2640 camera feed. A PIR motion sensor triggers capture to save power, and activity history is synced to cloud storage over WiFi.



Smart Water Intake Tracker

- A minimalist smart coaster that automatically detects drinking events and tracks daily water intake. A separate desktop display unit shows progress through a mechanical needle and RGB ring, with a reset button to restart tracking each day. Over time, the device learns the user's personal drinking patterns and sends reminders at moments when they are most likely to forget.

- Intended user: Anyone looking to build better hydration habits without manual logging or fixed reminder schedules.

- Technical elements: A load cell sensor detects cup weight changes to track drinking events, and cumulative intake is processed on the microcontroller. A lightweight ML model trained on the user's historical drinking patterns runs on-device to predict when a reminder is needed, replacing fixed-interval alerts with personalized nudges. Daily data is synced to cloud storage to continuously improve the model over time.